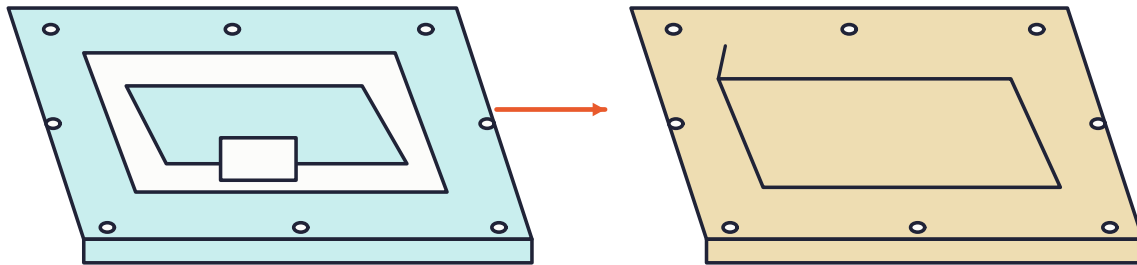


Make the soft parts

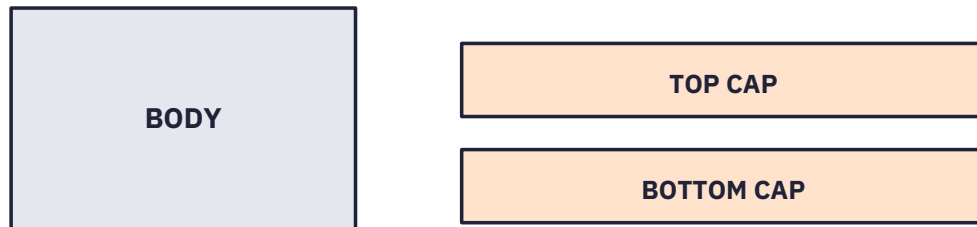
Silicone funnel + two foam caps + one body pour. A picture for each bench move.

FUNNEL



cavity + core + finished mandrel

THREE COLD-CORE POURS



1 Funnel: pages 2-16

Finish the printed tooling, prove the material stack, cast, open and inspect the removable funnel.

2 Cold core: pages 17-22

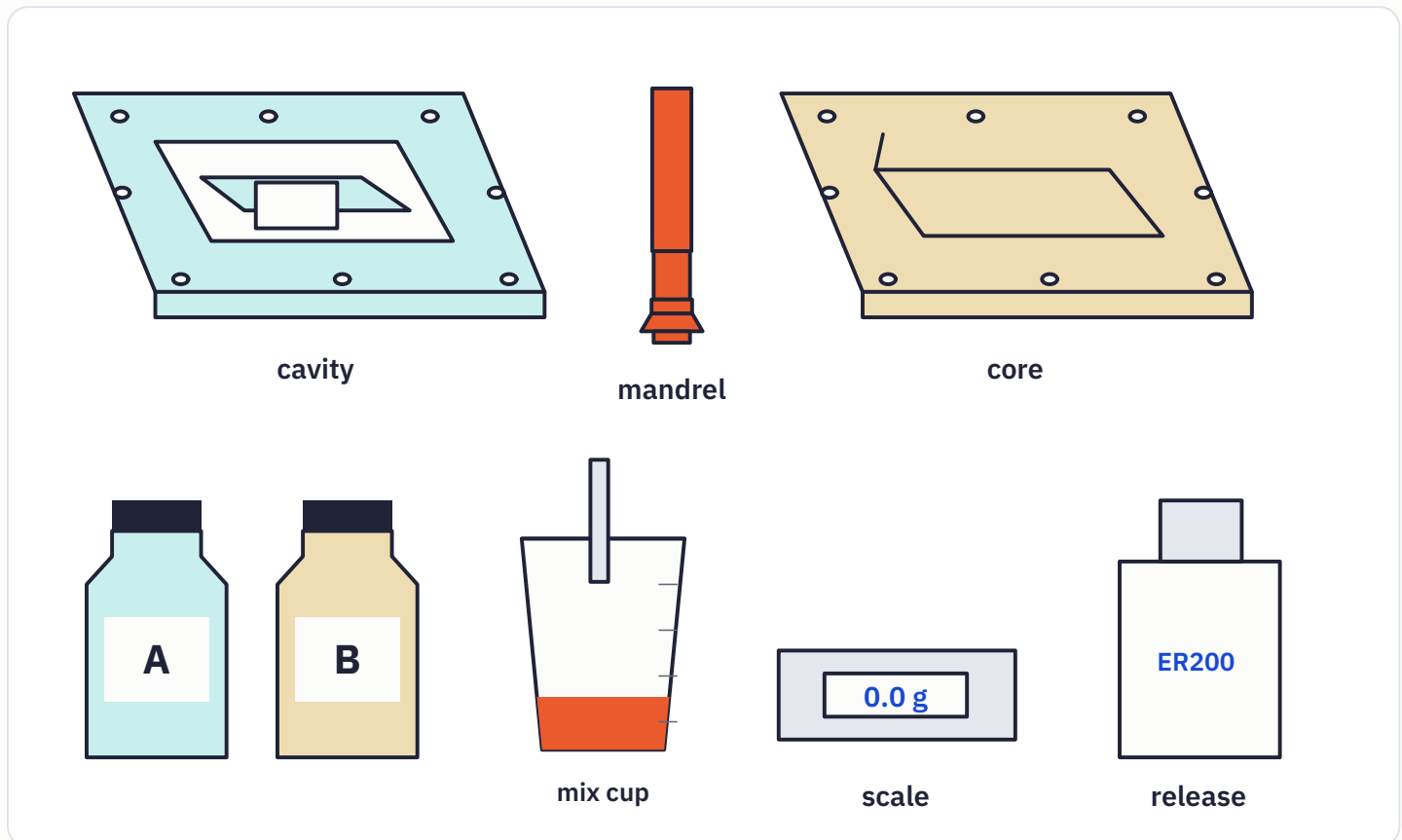
Prepare both cap pours and the body pour. Record the batch recipe before combining the foam liquids.

USE THIS AT THE BENCH

Coral marks the current move. Blue marks gauges, air paths and surfaces to keep clear. Pictures are schematic; named dimensions govern.

Stage the casting bench

Lay out the whole job before silicone A meets silicone B.



1 Tooling + closure

Cavity, core, finished mandrel, eight M4 x 20 bolts, 9 mm OD washers and nuts; two zip ties; proven sulfur-free entry-seal clay.

2 Liquids + handling

BBDINO 40A A/B, compatible black pigment, scale, cups, sticks, release, catch tray, vacuum chamber and pump.

3 Dry, ventilated work

Read the container instructions. Wear eye protection and appropriate gloves; keep release spray away from ignition sources.

READY

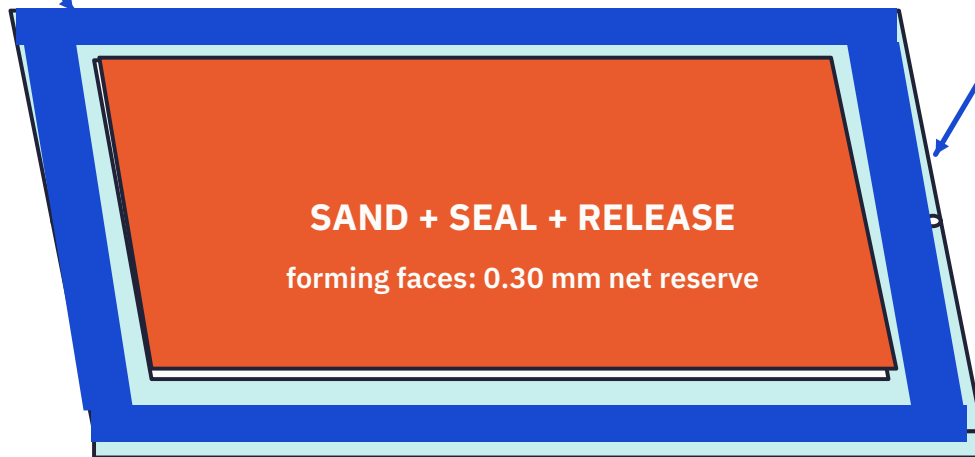
Room, materials and tooling are at the recorded batch temperature. The open mold and catch tray fit through the actual chamber opening.

Finish the forming faces

The coating allowance belongs to the forming surfaces. The parting lands remain bare.

mask bare lands

mask holes



also mask: locators, mandrel passage, V cradle and stop

1 Remove every support

Clear dry backs, V cradle and neck details. Remove loose plastic and smooth the forming slopes before applying the finish.

2 Measure a witness

Use the actual PETG, sanding and finish stack on a sample. Measure net growth; the shell reserve is 0.30 mm normal to the face.

3 Mask closure datums

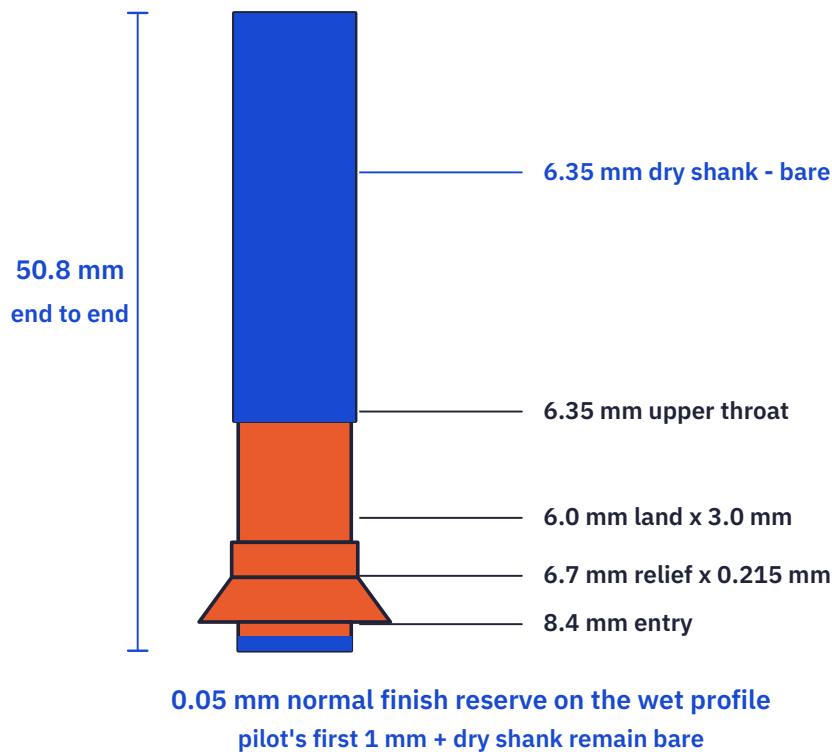
Keep lands, pegs/holes, bolt bores, mandrel passage, cradle and axial stop bare. Coated lands change closure height.

BEFORE CONTINUING

The finish is dry and coherent, and the coated halves still close on the bare lands. Coating compatibility and closure need a physical check.

Finish the small mandrel

The finished wet profile makes the funnel's staged sealing bore.



1 Finish to the reference

Use forming-mandrel-finished.step as the profile reference. Preserve the entry shoulder, tiny relief and cylindrical sealing land.

2 Mask both locating zones

The mandrel's wet reserve is 0.05 mm. Its first pilot millimetre and dry shank stay bare; the shell's 0.30 mm reserve is separate.

3 Inspect the thin relief

Check diameters and both relief edges under magnification. Finish the axial-stop end to 50.8 mm from the pilot end.

TOOL CHECK

Coating thickness is not a finished-profile measurement. The release force and coating retention of this finished printed mandrel remain unqualified.

Prove the material stack

A small same-stack witness protects the first full casting.

PETG



shell + tool

FINISH



sealer + release

CLAY



entry seal

actual 40A silicone + actual pigment on each actual contact stack

cure, then peel and inspect the contact face

1 Make the witness representative

Use the actual PETG, abrasion, sealer, release, clay and silicone batch. Include the planned pigment concentration.

2 Keep sulfur away

Platinum silicone can be inhibited by sulfur-bearing clay. Use the proven sulfur-free entry seal and clean handling tools.

3 Inspect after cure

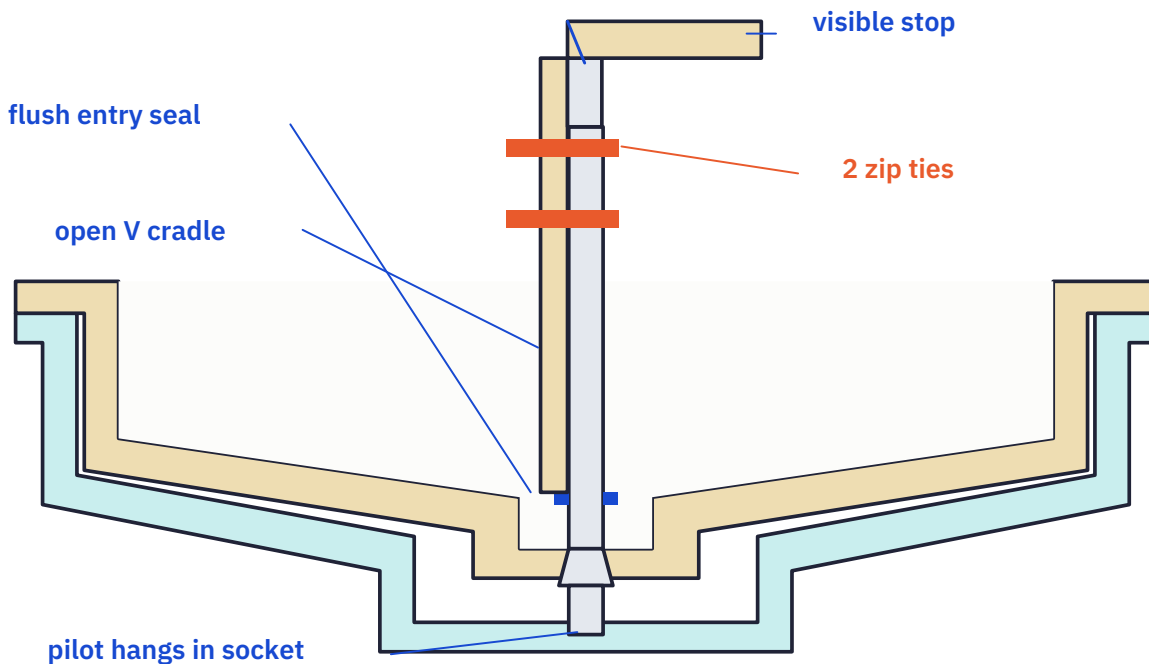
The rubber must cure at the contact face and release without tack, tearing or coating transfer. Record that result before the full cast.

HOLD IF GUMMY

A firm outer surface can hide inhibited rubber at the tool. Cure/release compatibility has no accepted full-tool record yet.

Seat the mandrel and seal

The V cradle and visible stop locate the tool; clay closes the loose annular passage.



SECTION / exaggerated wet features

1 Tie the bare shank

Pass the 6.35 mm dry shank through the 8.35 mm opening, seat in the V, meet the visible stop and secure with two zip ties.

2 Pack the entry seal

Pack proven removable clay around the mandrel at its forming-face entry. Smooth it flush; the illustrated seal is 2 mm deep.

3 Check the pilot

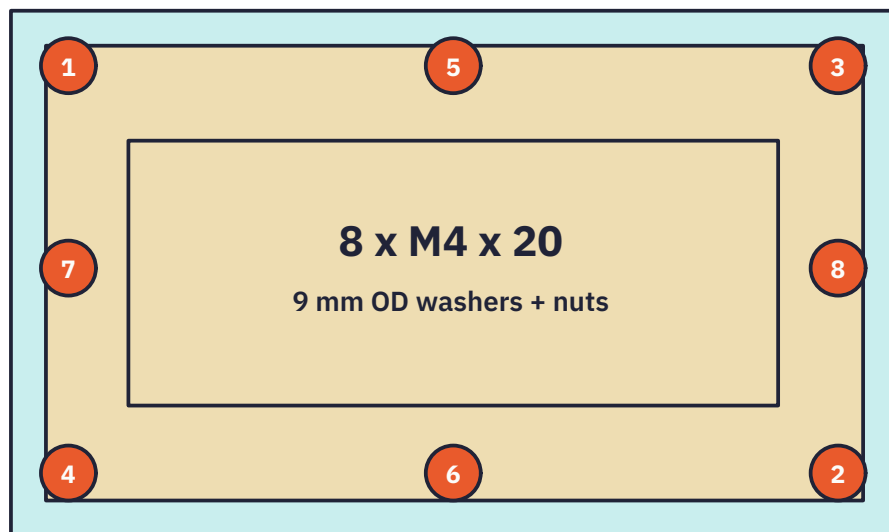
At nominal closure the pilot stands 1.5 mm below the block bottom in the 11 mm socket. It touches neither socket wall nor floor.

ALIGNMENT

The cradle locates the shank. Clay seals the passage. The finished wet profile must remain on the nominal bore axis.

Dry-close the mold

Bring the bare parting lands together before adding liquid.



opposite stations / small increments

bolt order schematic; actual asymmetric locators set orientation

1 Locate without force

The asymmetric short pegs establish drain orientation. One mating hole is slotted to accommodate spacing error.

2 Close opposite stations

Use eight bolts, washers and nuts or small clamps on the flat flange backs. Tighten incrementally until bare lands meet.

3 Backlight the seam

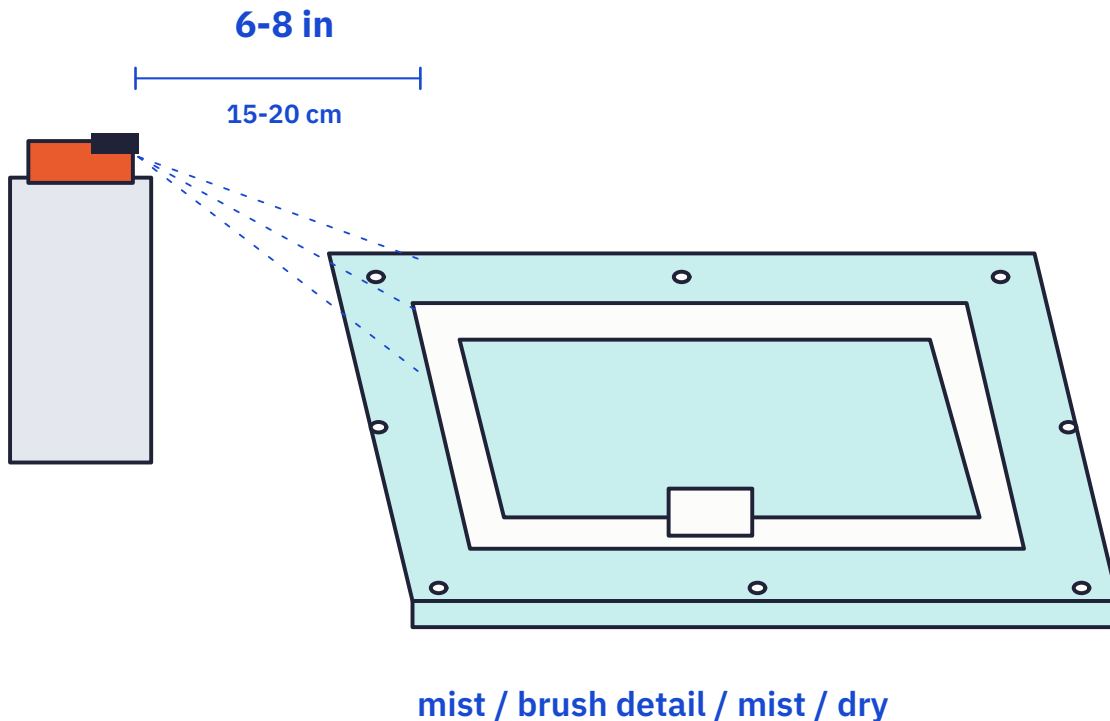
Check coated closure and mandrel clearance. Reopen for release and filling only after this dry fit succeeds.

HOLD IF IT WILL NOT CLOSE

A bowed or coated-open seam is not cured by adding silicone. Established stock-dowel and earlier shell fits do not qualify this finished tool.

Apply a light release film

A thin, dry, continuous film reaches the small wet features too.



1 Clean and shake

Use a suitable cleaner for the proven finish stack. Shake Ease Release 200 well; spray with ventilation and eye/skin protection.

2 Mist from 6-8 inches

Apply lightly, brush over fine details and add a second light mist. Cover the wet mandrel profile; avoid puddles and plugged relief.

3 Let it dry

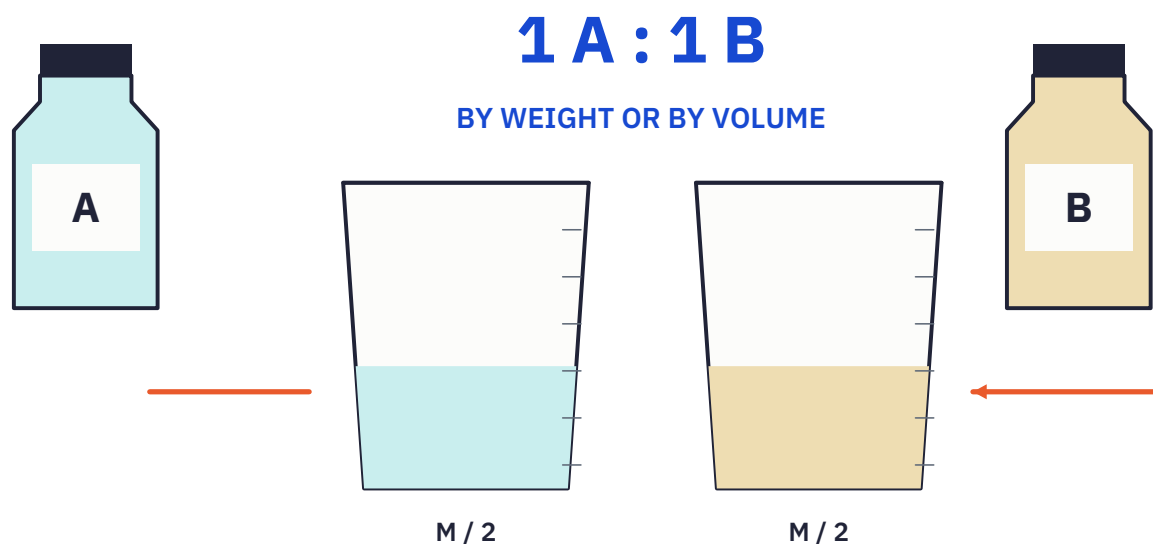
Recheck bare closure datums, entry seal and port openings. Apply a light fresh release film before each cast.

RELEASE CHECK

The same-stack witness must already have passed. Spray application does not establish compatibility, cleanability or finished food-contact acceptance.

Measure, color and mix

BBDINO 40A uses equal A and B. Choose one measurement basis for the whole batch.



219.3 mL nominal casting

about 248 g rubber + ports and mixing allowance

1 Set a batch quantity

Nominal geometry is 219.3 mL. The ~1.13 g/mL material estimate gives ~248 g in the casting; allow extra for fill ports and mixing loss.

2 Weigh the pigment

Use the actual BBDINO platinum-compatible black. Project limit: at most 2% by weight. Include it in the same-stack cure/release witness.

3 Mix the measured components

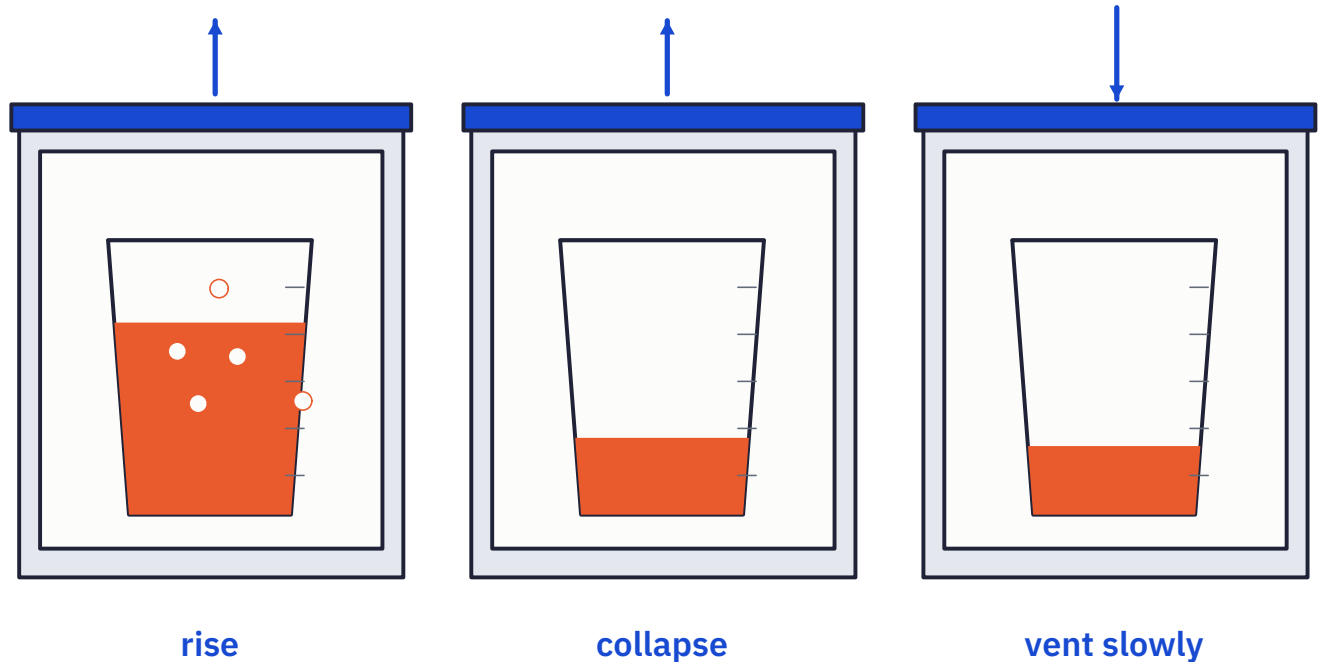
Scrape the cup wall and bottom until uniform. Start the working-time clock when A and B meet; no unmixed streaks remain.

BATCH CLOCK

Maker working time: 30 min at 23 C. Record room and liquid temperature; warmer conditions shorten the window. Follow the supplied batch instructions.

Degas the mixed silicone

This 40A grade requires degassing. Give the rising foam room above the liquid.



1 Choose an ample cup

Mixed silicone rises during evacuation. Use a larger container or split the batch if the available cup cannot contain the froth.

2 Evacuate under control

Degas in a separate container with expansion space. Observe the rise and collapse; control the valve to prevent overflow.

3 Vent and pour while fluid

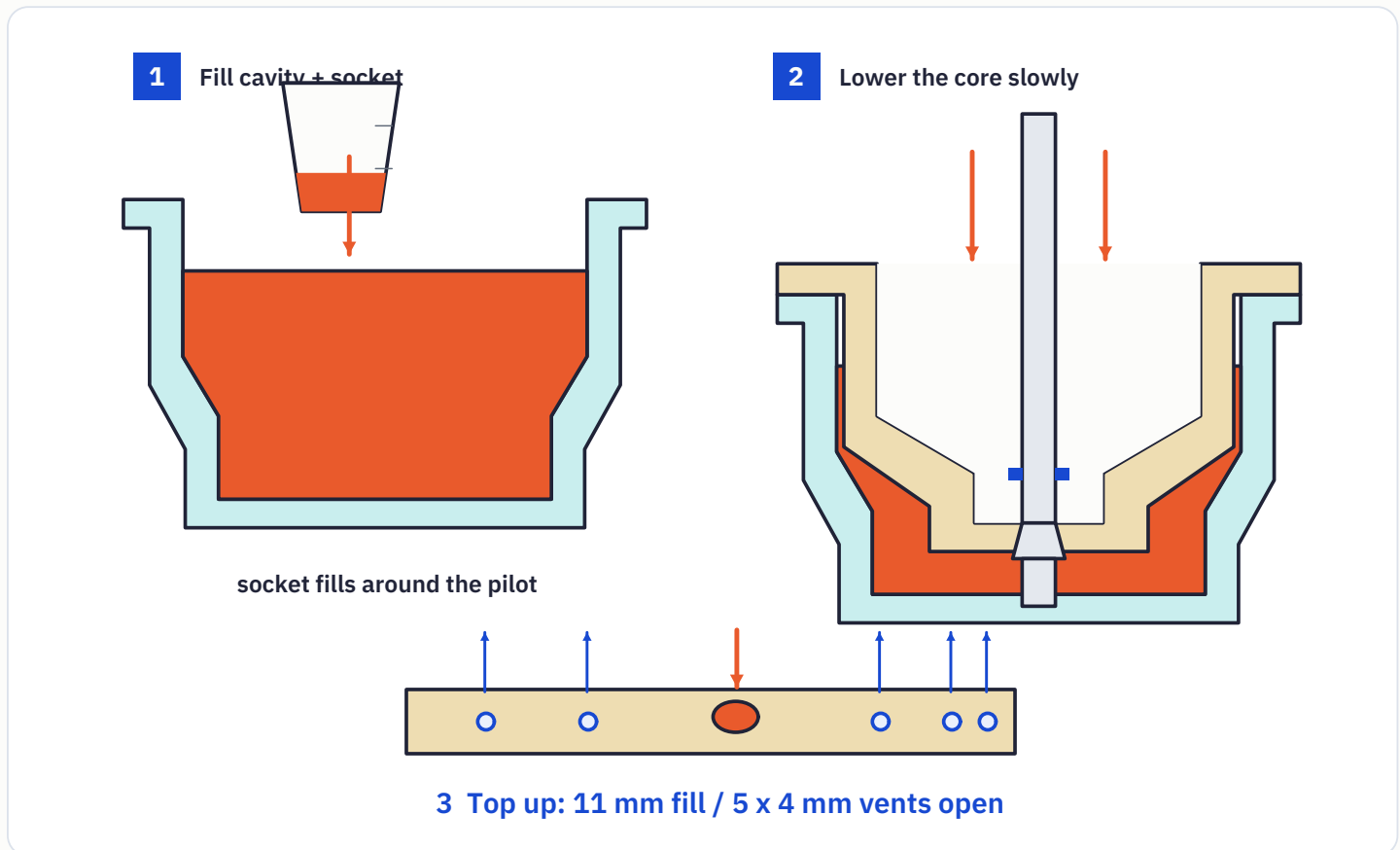
Vent slowly, remove the cup and continue the casting sequence within the measured working-time window.

USE THE ACTUAL EQUIPMENT INSTRUCTIONS

The guide supplies no universal vacuum setpoint or timer. Follow the chamber/pump and silicone batch instructions; watch the material.

Fill, lower, then top up

Fill the open cavity and rod socket before the core closes the narrow gap.



1 Wet the entire open cavity

Include the rod socket and drain block. The degassed mix must still flow when the finished mandrel and core enter.

2 Seat the core evenly

Lower slowly with the mandrel installed. Close and hold the flanges on their bare lands without sealing the vents.

3 Top up through the fill hole

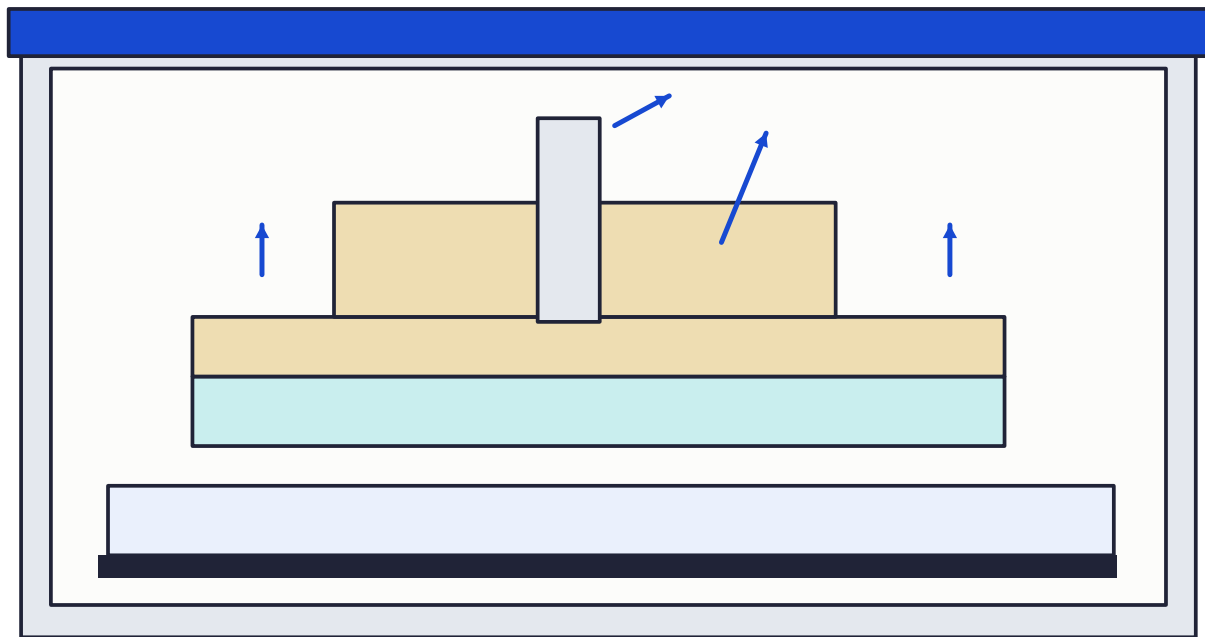
Use the 11 mm fill opening. The five 4 mm vents remain clear for air; catch overflow and keep the dry backs open.

CHECK

The parting lands remain closed, the pilot is aligned, and liquid reaches the small block and thin ramp gap. No pressure injection.

Cycle the filled mold gently

If using a filled-mold vacuum cycle, the whole tool shares the chamber pressure.



fill + vents + both dry backs open

catch tray / overflow clear of pressure paths

1 Keep openings connected

Place the complete mold inside the chamber. Fill, vents and dry backs must communicate with chamber air; overflow cannot seal them.

2 Evacuate and vent slowly

Perform any cycle while silicone is fluid. Observe tool/overflow behavior, vent gently, then recheck and top up the fill level.

3 Return to ambient for cure

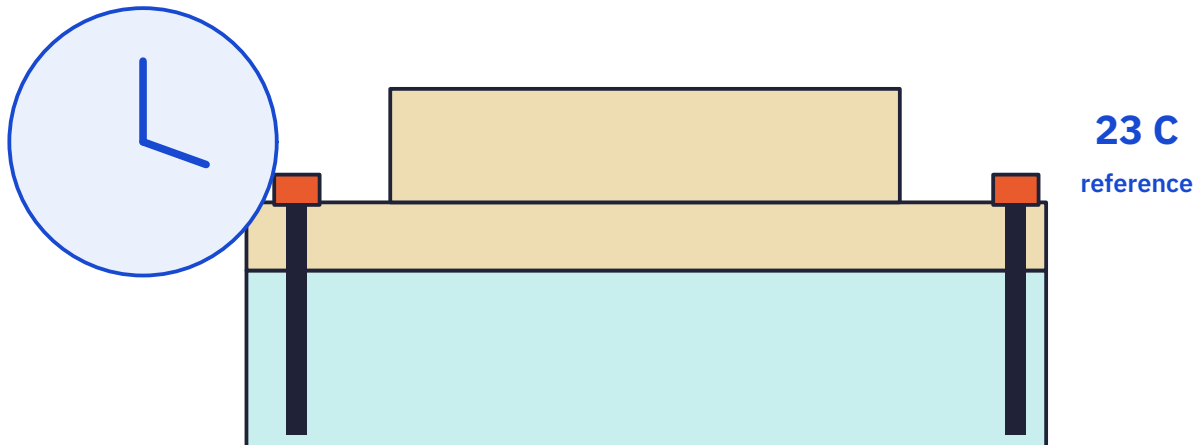
Keep the flanges held together through room-temperature cure. The tooling is not rated for a sealed atmosphere differential.

PHYSICAL LIMIT

Coated closure and a vacuum cycle are not physically qualified for this exact tool. Stop on seam opening, distortion or a blocked pressure path.

Hold the flanges through cure

Leave the tool undisturbed at ambient pressure while the silicone develops its cure.



5 h project demold hold

24 h project full-use hold

also inspect a same-batch witness

1 Log time and temperature

Record A/B mixing start, pour finish, room/liquid temperature and batch identity. Warmer and cooler shops change the cure window.

2 Use the conservative project hold

Project material record uses 5 h before demold at 23 C and 24 h before full use. Check the supplied batch insert and cured witness.

3 Keep the tool assembled

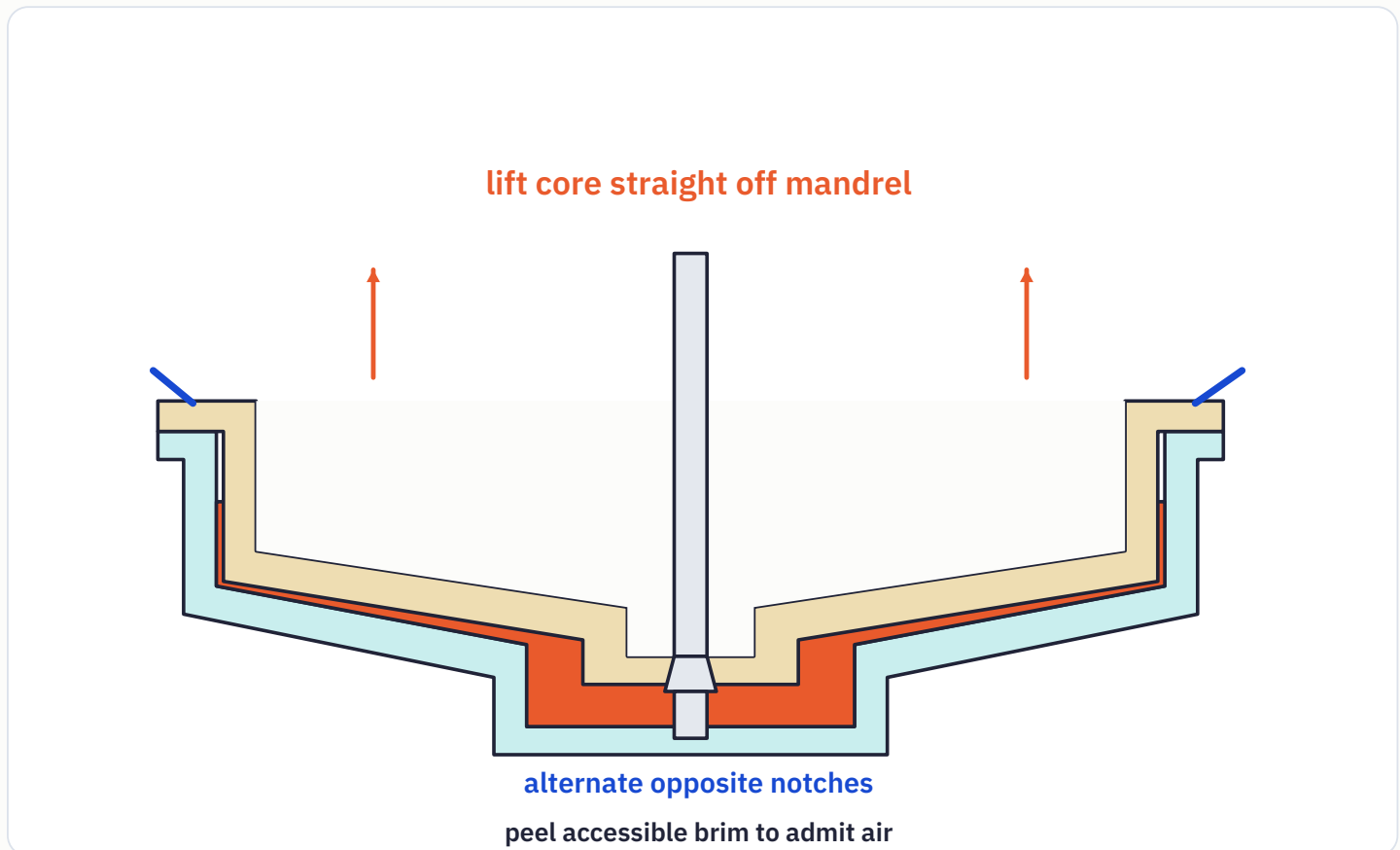
Maintain ambient pressure and closed flanges. Do not pull the thin brim or mandrel until the contact surfaces have cured.

CURE IS NOT FOOD-CONTACT RELEASE

The manufacturer's direct page lists 3 h; the project holds 5 h conservatively. No qualified post-cure bake schedule is established.

Open in small movements

Free the core first; keep the casting and mandrel together while peeling from the cavity.



1 Free the small restraints

Trim overflow at port mouths. Cut the two mandrel zip ties and remove the entry-seal clay before separating the tooling.

2 Alternate opening points

Use a blunt tool at opposing edge notches, moving a little at a time. Peel accessible silicone brim to let air enter.

3 Lift, then peel

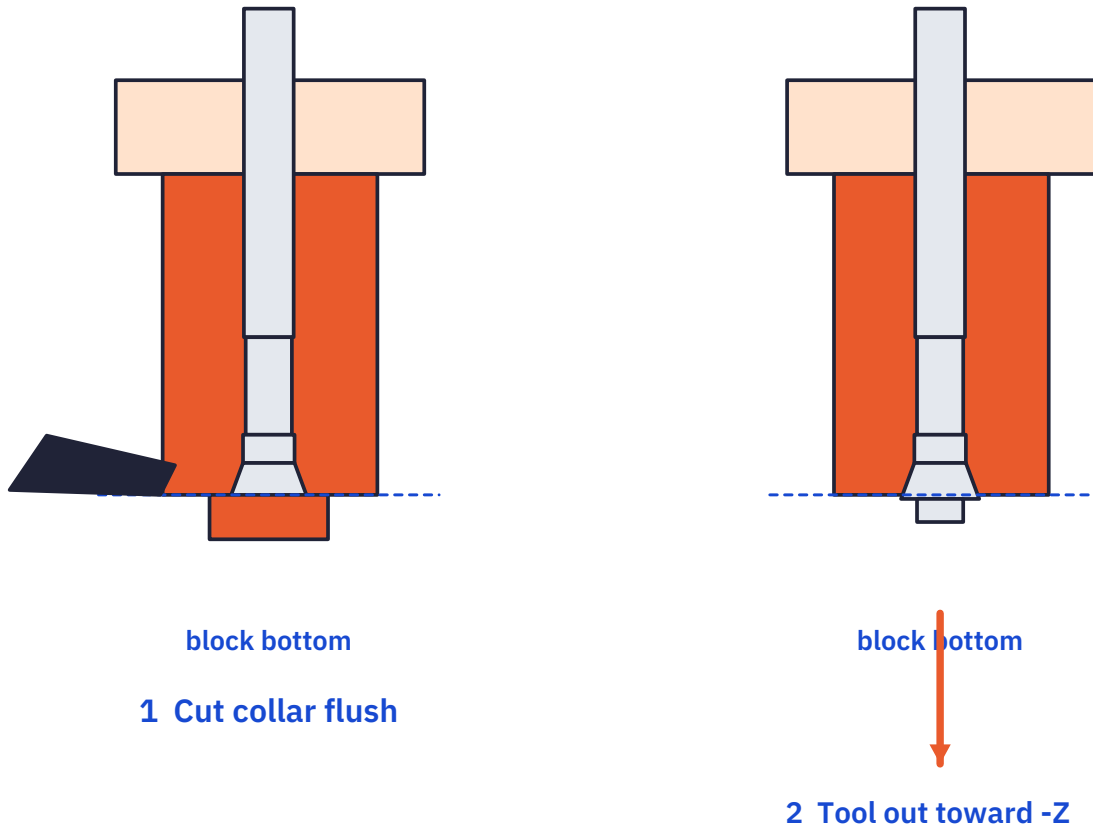
Lift the core straight off the mandrel. Peel the casting and mandrel together from the cavity; keep the drain block supported.

STOP ON STICKING

Do not yank the thin brim or use the mandrel as a lever. Release force and surface damage are first-casting observations to record.

Trim the collar, then withdraw

The socket makes a sacrificial silicone collar. Remove it before the enlarged entry passes through.



1 Support the drain block

Trim only the socket collar to the flat block-bottom face. Keep the final bore and 6.0 mm sealing land intact.

2 Withdraw toward the entry

Move the mandrel toward -Z after trimming. The 6.35 mm shank passes the 6.0 mm land by elastic silicone expansion.

3 Inspect both surfaces

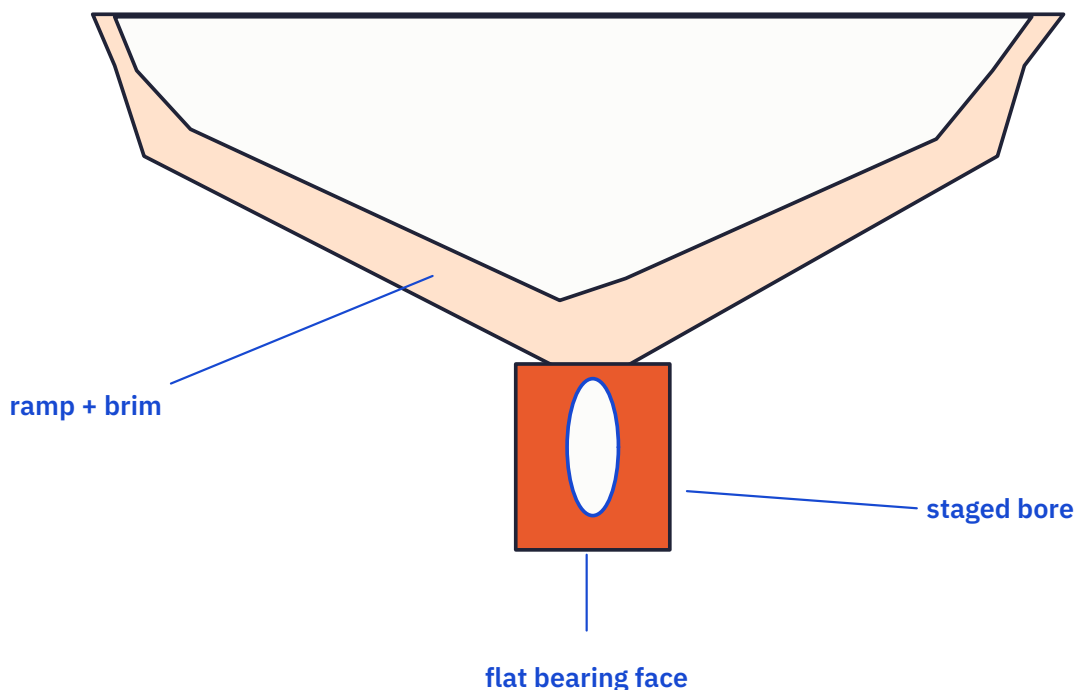
Check the sealing land, entry shoulder, thin relief and tool coating for tears, transfer or damage before another cast.

RELEASE REMAINS A TRIAL

The nominal diameter change is 5.83%; it does not establish acceptable extraction force, tearing resistance or tooling life.

Inspect and clean the funnel

A complete casting has a continuous ramp, an open staged bore and an intact bearing face.



1 Inspect every wetted surface

Look for pinholes, bubbles, tack, tears and coating/clay transfer. Trim port overflow without cutting into the nominal brim or drain block.

2 Clean with the qualified process

Release traces and residue must be removed. The finished silicone, black pigment and post-process need the wetted-surface qualification.

3 Check its actual installation

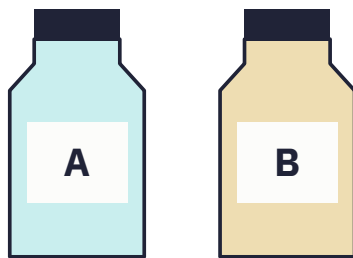
The flat block face seats on the frame; the bore engages the drain elbow. Record seating, removable lift-out and leak behavior.

HOLD FROM FOOD SERVICE

No qualified bake schedule or finished-mixture food-contact result is recorded. A cured, attractive part alone does not close that gate.

Prepare the three foam pours

Two cap cavities and one body cavity use the acquired FSD 2 lb closed-cell PU foam.



BATCH RECIPE

ratio + measurement basis

liquid / room temperature

mix + working window

A/B quantity for each shot

release / cure / trim time

TOP / BOTTOM / BODY

1 Confirm the actual batch

Use FSD B08R7TX8QJ and its container instructions/SDS. The official page says equal parts; ratio basis and process window need the batch record.

2 Record each shot

Top and bottom caps differ. Set the measured A/B quantities, expansion allowance, mixing/working time and cure/trim hold before combining.

3 Stage a dry, ventilated bench

Use dry cups/tools, protective gloves and chemical eye protection. Keep moisture out of the liquids and vapors away from the operator.

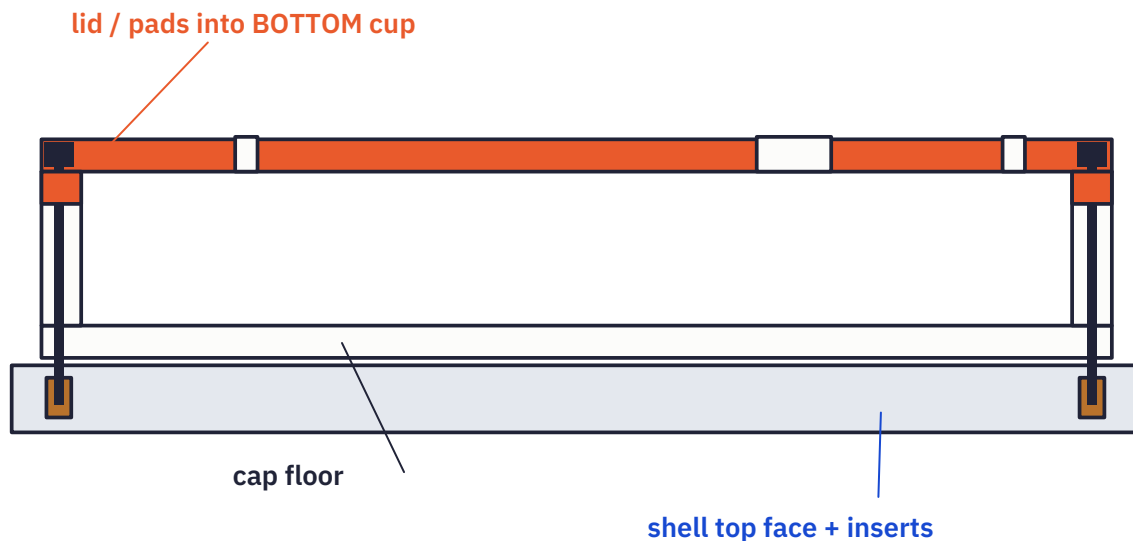
HOLD MIXING AND POURING

The repository has no calibrated cap/body shot recipe or verified batch timing. Geometry and nominal density do not supply liquid dose or cure time.

Clamp each cap mouth-up

Use the shell's top face as the pour fixture for both cap stacks, one after the other.

10 x M3 x 25 through lid + cap



TOP: flat lid underside / 4 deck inserts first

BOTTOM: counterbore pads face into the cup

1 Label TOP and BOTTOM

The top cap carries conduits and four deck-mount columns. Press the four full-length RX-M3x5.7 inserts into those columns before pouring.

2 Use installed shell threads

Ten full-length M3 inserts in the shell's top face supply the pour-clamp threads. Lid and cap have clearance holes.

3 Build the correct stack

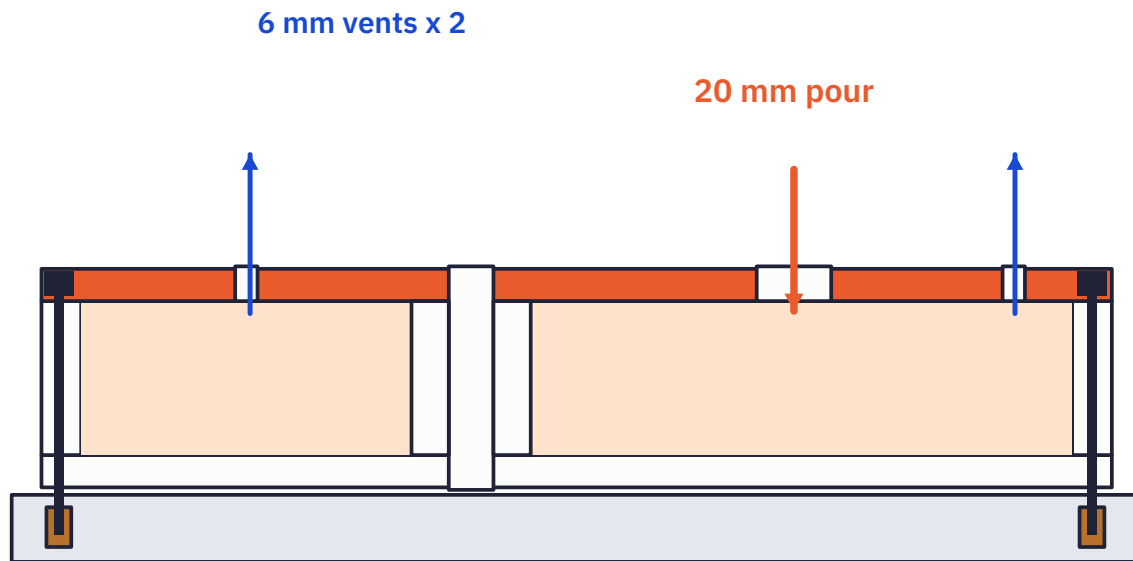
Cap floor down on the shell. Bottom lid pads enter the cup; top lid underside is flat. Close with ten M3 x 25 screws.

CHECK

The lid is held by the complete screw pattern, not its own threads. All top-cap conduits and both vent paths remain open.

Pour the two cap cavities

With the recorded foam recipe ready, liquid enters the lid and air leaves through both vents.



TOP 12.6 mm / BOTTOM 16 mm pour depth

top-cap section / one stack at a time

1 Mix the recorded shot

Use the batch recipe's measurement basis, A/B quantities, temperature and working window. Have the clamped cap ready before mixing.

2 Pour into the 20 mm hole

Keep the two 6 mm vents clear while foam expands. The lid stays clamped; do not cover the vent paths to force fill.

3 Cure, trim, unbolt, repeat

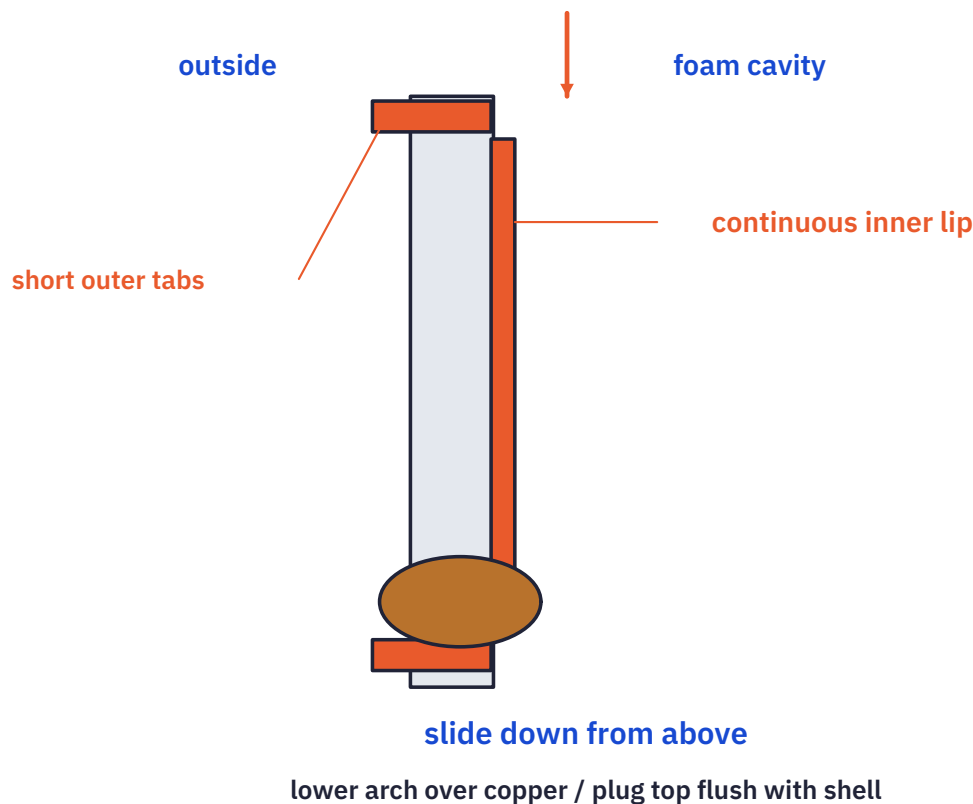
After the recorded cure/trim hold, trim pour/vent overflow to the plate. Remove ten screws and repeat for the other labeled cap.

PRESERVE THE TOP FEATURES

Trim flush without cutting a valve cradle, conduit, anchor or deck-mount feature. No numeric shot size or cure timer is implied by this page.

Close the slots; protect access

Everything foam will bury is installed, routed and proved before the body pour.



1 Prove the buried work

Carbonator and inlet jet qualified; coil/probes/reeds and fitted fluid lines checked. Tug-test inaccessible push fittings before encapsulating.

2 Seat both copper plugs

Continuous lip faces the foam cavity. Shell wall lies between that lip and the short outside tabs. Lower arch seats over the copper.

3 Keep service paths open

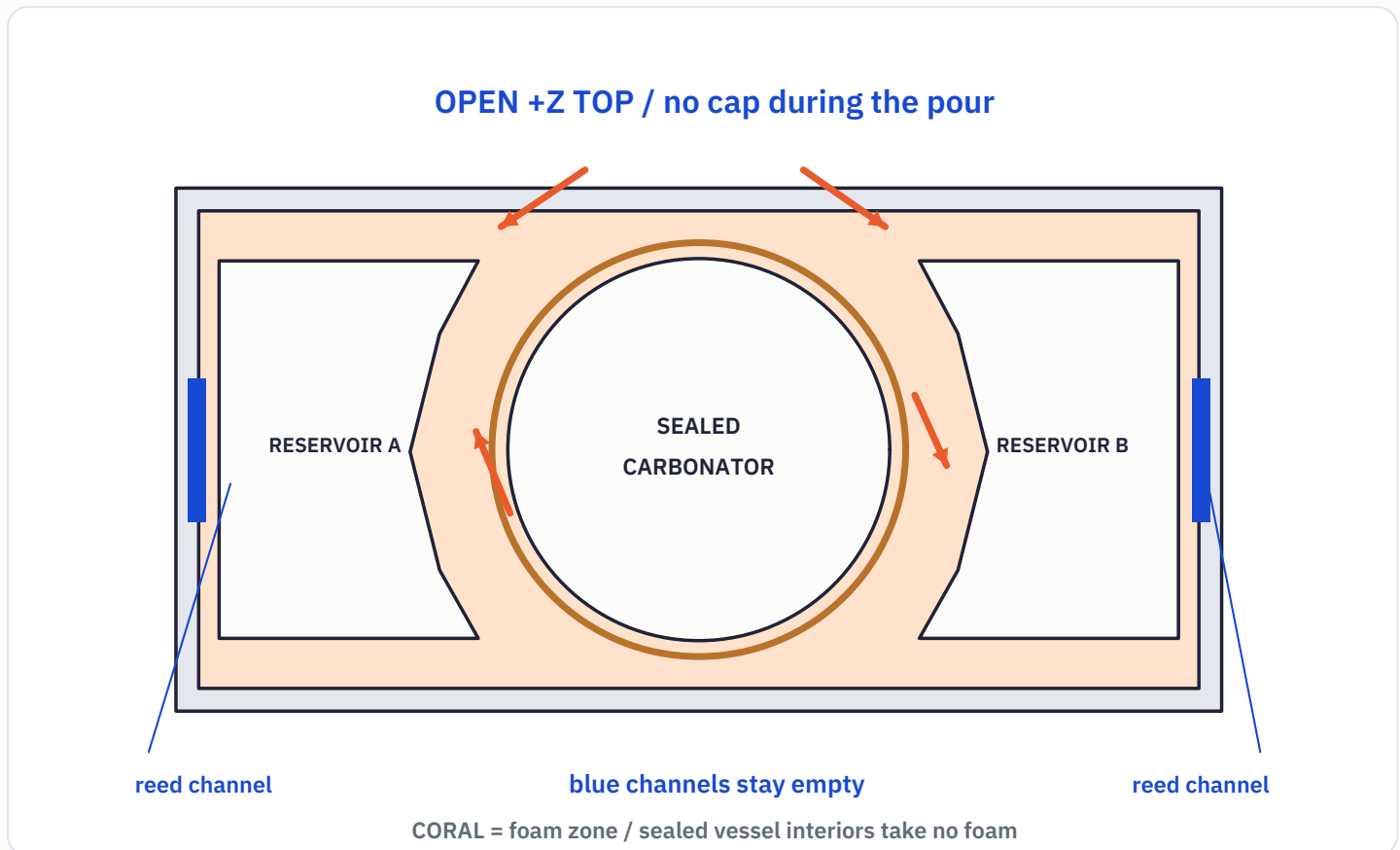
Protect reservoir vent/fill openings, top conduits, relief outlet and both reed channels. Reed columns are installed after body foam cure.

PROBE AND ACCESS CHECK

Both probes and lead entries need foam coverage without a trapped void. Do not bury an unproved sensor, line, relief path or fitting.

Pour through the open top

The body foam fills around the sealed carbonator and reservoirs; its connected zone reaches the coil.



1 Mix the recorded body shot

Use the batch record's quantities, temperature and mix/working window. The top remains open; cap stacks are already poured separately.

2 Pour directly into the body

One pour fills the connected outer gap and space around the carbonator/coil, through the open paths at the reservoir-pocket ends.

3 Watch fill and squeeze-out

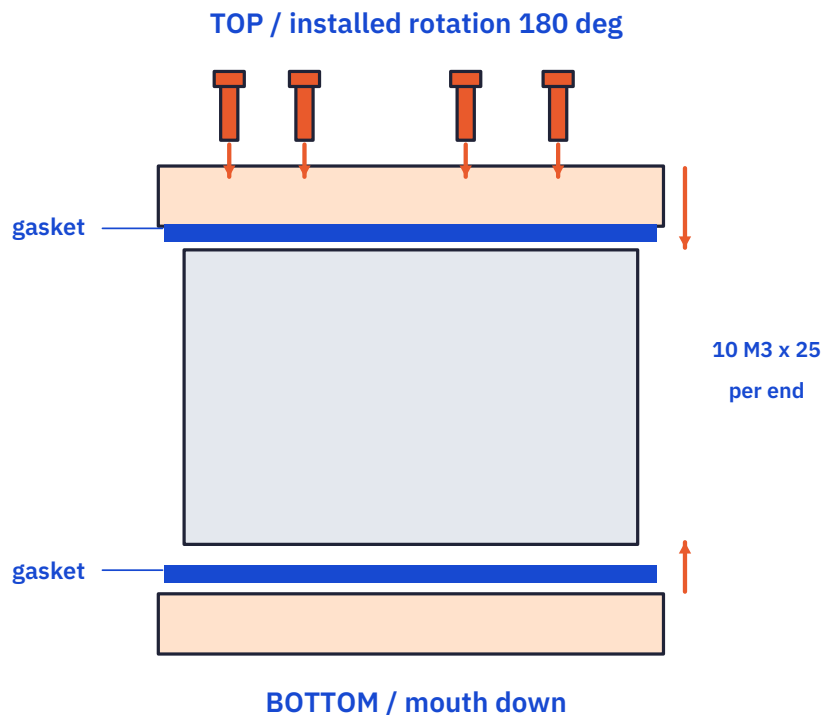
Preserve empty reservoir/reed cavities and open tube paths. Foam may emerge at tube exits; leave trimming until the recorded cure hold.

DO NOT ASSUME HIDDEN WET-OUT

Probe voids and trapped gaps around the embedded coil are insulation defects. A risen top surface does not prove the hidden zone is full.

Trim and close the cold core

All three pours are cured before the gaskets and caps complete the insulated stack.



1 Trim only cured overflow

Use the qualified trim method. Protect tubes/leads, plug webs, cradles and insert hosts; clear all conduits, vents and reed channels.

2 Install reeds, gaskets and caps

Drop reservoir reed columns in, route their cables out the top, and lay both TPU gaskets. Top cap rotates 180 deg; bottom mouth faces down.

3 Close and inspect

Ten M3 x 25 screws per end. Identify top orientation by deck/cradle features. Check open paths, gasket seating and recessed screw heads.

FINISHED

Both caps and body cured; exposed foam flush; probes protected; channels and relief path open. Buried-fill quality needs its own process evidence.